BIOHACKATHON EUROPE 2026 · PROJECT 9: FROM ALIGNMENT TO INSIGHT

Interface explorer

Proteins work by touching each other. We mapped where they touch onto the human genome, and disease-causing mutations pile up exactly there.

Try it: https://cmdcolin.github.io/interface-explorer/
Progress as of 11 November 2026

THE WHOLE THING ON ONE SLIDE

TL;DR: a mutation where two proteins touch is more likely to cause disease, and now you can look each one up

THE IDEA

handshakes

Most proteins do their job in pairs. Change one building block on the surface where they meet and the handshake can fail.

WHAT WE DID

5,350 proteins

For every human disease protein, we lined up three things: its contact surfaces in 3D, the DNA that codes for them, and the variants seen in patients.

WHAT WE FOUND

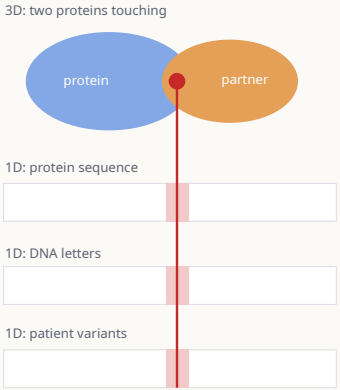
2.4× as often

Harmful variants sit on a contact surface 2.4 times as often as harmless ones. Another 23,233 variants that nobody can yet classify sit there too.

The contact surfaces come from AlphaFold's predicted protein pairs and, for the team's receptor set, from lab-solved structures. Everything is on a public website with a genome browser and a 3D viewer.

HOW IT FITS “FROM ALIGNMENT TO INSIGHT”

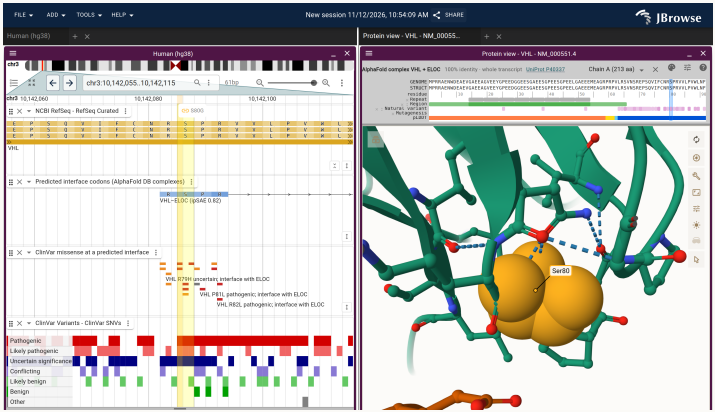
TL;DR: the project locks the 1D alignment and the 3D view together; this does the same with the genome as the 1D view



The project asks for	What the explorer does
1D and 3D locked in sync	genome browser and 3D viewer linked, side by side
Interaction surfaces that stand out	contact surfaces drawn as tracks for 1,331 disease genes
GPCRs as a demo family	the team’s 45 receptor structures, G-protein contacts mapped
No structural expertise required	look a gene up by name, click a variant
Conservation on surfaces, from an alignment	not here: no multiple sequence alignment view

WHAT IT LOOKS LIKE

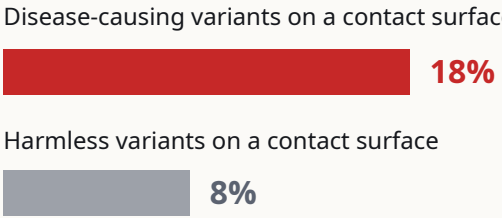
TL;DR: one example. A known cancer-syndrome mutation sits right where VHL grips its partner



- **VHL** guards against tumours; inherited faults cause von Hippel–Lindau syndrome
- 27 of its 213 building blocks touch its partner; **48 harmful variants** sit on them
- Shown: one of them, S80G, on the gene (left) and in 3D (right). 29 more on that surface are unclassified

THE MAIN RESULT

TL;DR: harmful variants are about twice as likely to sit where proteins touch

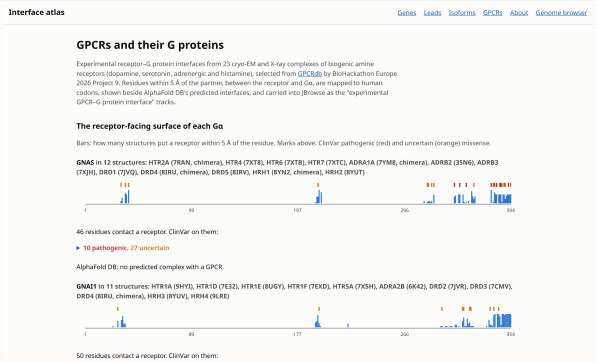


2,260 of 12,655 pathogenic and 677 of 8,247 benign ClinVar missense variants, in 906 proteins that have both.

- Compared fairly, protein by protein, the odds are **2.4×** (between 2.0 and 2.8)
- It is not just “the mutation is in a solid part of the protein”: on the folded surface alone it is **3.2×**
- So a contact surface is a reason to look harder at a variant nobody has classified yet

USING TEAM DATASET 1

TL;DR: on the team's receptors, AlphaFold's predicted contact surfaces match the lab's



- 45 lab-solved receptor structures picked by the team; 23 include the G protein the receptor signals through
- Where both exist, the predicted surface covers **87–90%** of the experimental one
- **10 known disease mutations** sit where the Gs protein docks to receptors; AlphaFold DB has no prediction there

GENE VERSIONS (ISOFORMS)

TL;DR: one gene makes several versions of a protein, and some versions lose a contact surface

- We lined up 7,648 alternative versions against the standard one
- **290** lose half or more of a contact surface
- **2,123** lack a position where a disease mutation is known
- Example: the brain's version of APP, the Alzheimer's protein, lacks the part that binds three protease partners



HONEST LIMITS AND NEXT STEPS

TL;DR: these are leads, not proof, and there is room to take it further

Limits

- Most contact surfaces are AI predictions, not measurements
- Only 1 in 4 disease proteins has a predicted partner
- A variant on a contact surface is a hint about how it does harm, not a diagnosis

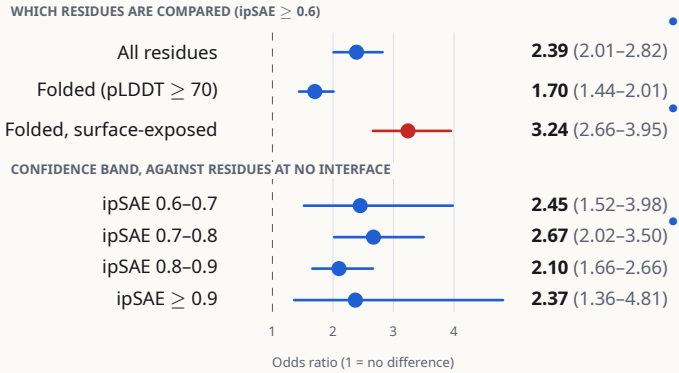
Open for the team

- 675 of the 23,233 unclassified variants share a position with a known harmful one: the Leads page lists them. Which first?
- The 22 receptor structures without a G protein are not analysed yet
- The contact-surface lists are plain JSON and bigBed files: could Mol*, iCn3D or Jalview show them too?

BACKUP · THE STATISTICS

Pathogenic variants sit at predicted interfaces 2.4× as often as benign ones

Within-protein comparison (Mantel–Haenszel odds ratio, bootstrap 95% CI) over the 906 proteins with both pathogenic and benign missense variants.



2.39 (2.01–2.82)

1.70 (1.44–2.01)

3.24 (2.66–3.95)

2.45 (1.52–3.98)

2.67 (2.02–3.50)

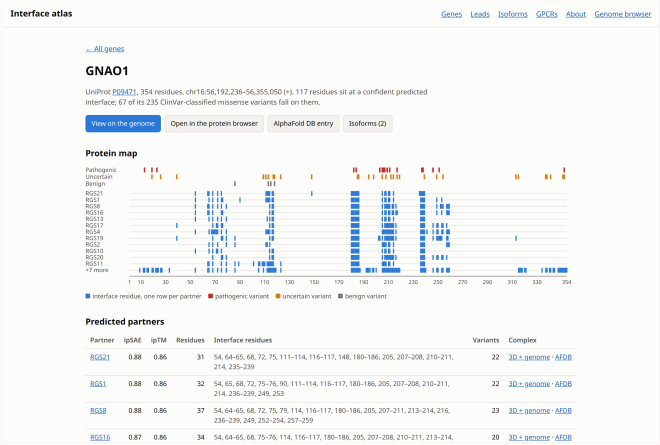
2.10 (1.66–2.66)

2.37 (1.36–4.81)

- It is **not just “folded vs disordered”**: among folded, surface-exposed residues the odds ratio is 3.2 (2.7–4.0)
- The signal **does not depend on confidence**: every ipSAE band from 0.6 up sits between 2.1 and 2.7, with overlapping intervals
- It **adds to AlphaMissense**: among variants AlphaMissense calls likely benign the odds ratio is still 1.4 (1.1–1.9), and among those it calls likely pathogenic 1.4 (1.0–2.0)

BACKUP · GENE PAGES

From a gene to its partners and its interface variants



- **GNAO1:** pathogenic variants at the Gβ interface (L13P, I19M, L23F) and the RGS interface (T182I, Q205P, R206)
- **VHL:** 48 pathogenic variants at the elongin C interface
- The 23,233 variants of uncertain significance at an interface are the leads
- Tracks carry one feature per partner, e.g. “GNAO1–GNB2 (ipSAE 0.73)”

BACKUP · ENGINEERING

Reproducible, tested, and deployed on every push

When	Milestone
10 Nov	Prototype: AlphaFold DB interfaces and ClinVar on hg38; website around JBrowse
11 Nov, am	Every disease gene; ClinVar from NCBI; isoform browser
11 Nov, pm	Isoforms for all 3,423 genes; GPCR page from the team's selection, with chimeric Gα
11 Nov, eve	Pipeline ported from Python to TypeScript; type check and tests gate the deploy

- **Seven pipeline steps** in TypeScript, run directly by Node 24 with no build step: `pnpm all`
- Built on **g2p_mapper** and **p2s_mapper**, the libraries behind the JBrowse protein3d plugin
- The port reproduces the Python outputs: all four BED files byte for byte
- **89 unit tests** in 13 files
- GitHub Actions deploys the static site with JBrowse Web to Pages